



SafeAlert — Documentation

Location-aware emergency alerts for Ghana: one app for staying informed, calling for help, and reporting danger to your community.

[Open the live app](#)[Admin portal](#)[Source \(private repo\)](#)

OVERVIEW

Three jobs, one app

Stay informed

A single feed merging UN global disaster alerts (GDACS), US weather service alerts, official admin bulletins, community reports, and broadcasts — plus live weather, all centered on where you are.

Call for help

Hold-to-send SOS: your emergency contacts get a full-screen siren alarm in-app, an automatic SMS with your live location, and a phone call — within seconds.

Report danger

Anyone can post a missing-person, flood, fire, or accident report — with a photo — that reaches every user's feed and map. Admins publish official bulletins and broadcasts.

The app is an installable **PWA** (open the link, "Add to Home Screen") and also runs natively via Expo. The default area is **KNUST, Kumasi** until GPS resolves. It self-updates on each launch.

USER GUIDE

Getting started

- 1 Open kuwguap.github.io/safealert and add it to your Home Screen (Share → **Add to Home Screen**).
- 2 **Sign up** with name, email, and password. Add **your phone number** (receives SMS for Extreme AMBER alerts) and at least one **emergency contact** — include their SafeAlert email so they get the in-app siren when you SOS.
- 3 Allow **location access**. Until granted, the app shows the KNUST default area with a tap-to-retry banner.

Accounts work on any device. *Forgot password?* emails a reset link; passwords can also be changed in Settings.

Home (Alerts tab)

- **Location pill** — your position, labeled by the closest landmark ("Near KNUST Students' Clinic"). Tap to switch to a saved place.
- **Weather card** — current conditions and the next hours, for the active location.
- **Alert feed** — color-coded cards (amber = AMBER, blue = flood) with distance from you. Tap for the full detail: description, attached photo, affected-area map, and actions.
- **"+" button** — post a community alert (missing person, flooding, fire, accident, other) with an optional photo. Reaches everyone within ~20 seconds.
- **Pull down to refresh.**

Places & Settings

- **Places** — search any Ghanaian town, suburb, or junction (Ghana-first search), save it, set it *active*: weather, alerts, and maps all re-center on it. Watch over family elsewhere.
- **Settings** — account & contacts, change password, alert-type filters, °C/°F, default map layer, data-source status, and (for admins) the dashboard button.

EMERGENCY

The SOS system

Hold the red SOS button for ~1.5 s — the progress ring prevents accidental triggers. On completion, three things happen at once:

- 1 **In-app alarm.** Contacts on SafeAlert get a full-screen red alarm with a looping siren and vibration that continues until they press *Stop alarm* — with one-tap "Open live location in Maps". A stopped alarm never re-fires, and an SOS from the last 10 minutes still alarms if their app was closed.
- 2 **Automatic SMS.** Every contact is texted your live-location link through the Arkesel gateway (works on any Ghanaian number). If the gateway is unreachable, your SMS app opens prefilled instead — the status line tells you which happened.
- 3 **Call & resend.** Buttons appear to call your first contact and resend the SMS.

- **"I'm Safe" check-in** — one tap tells your contacts (and the admin log) you're okay.
- **Ghana quick dial** — Police/DOVSU **191** · Missing Children Ghana **059 459 4662** · NADMO **122** · Poison Control (GPCC) **020 222 2174**.
- **Incoming SOS preview** — every account can preview the alarm without alerting anyone.

MAPS

Three layers, live data

LAYER	WHAT IT IS
Map	Street map (CARTO Voyager) — the default; crisp roads and names with excellent Ghana coverage.
Satellite	Esri imagery with a place/road label overlay.
3D	Real WebGL (MapLibre GL): extruded 3D buildings on a dark basemap, cinematic fly-in, slow orbit until touched; drag, pinch, two-finger rotate and tilt.

All layers show **landmark markers** — schools (purple), hospitals (coral), police (blue), fire (amber) — with a color legend, neighborhood labels, your pulsing beacon, and + / - / © controls. Landmark data streams live from OpenStreetMap as you pan.

ADMIN

Admin dashboard /admin

Open from Settings or directly at /safealert/admin. The admin role is verified server-side.

CARD	WHAT IT DOES
Publish alert	Official bulletins: AMBER / Flood / Severe Weather / Earthquake / Wildfire, headline, details, optional photo, severity. Reaches every device in ~20 s. AMBER + Extreme additionally SMS-blasts every registered user phone.
Broadcast	Title + message pushed to all devices as a toast, notification, and feed card.
Bulletins list	Everything published, with per-item Remove (removes for everyone).
Incoming activity	Live log of SOS, "I'm Safe" check-ins, and sighting tips — user, location, time.
DB status	Supabase reachability and latency, in the header pill.
Smoke tests	One-tap checks with ✓/X results: Notification · SOS alert (fires the real alarm at your account) · Broadcast · Backend ping · Refresh feed · Clear bulletins · Clear activity.

UNDER THE HOOD

Architecture

CLIENT — EXPO / REACT NATIVE + TYPESCRIPT Screens (Home · Map · Places · Settings · Emergency · Admin · Post · Auth) · state (AuthContext, AppContext) · MapPanel & Map3D · SosAlarm · image picker

SUPABASE

Auth (JWT) · Postgres + RLS:

`profiles` ,

`safealert_bulletins` ,

`safealert_events` · Storage

`safealert-media` · Edge

Function `send-sms`

PUBLIC DATA (KEYLESS)

GDACS · US NWS · Open-Meteo ·

OSM Overpass & Nominatim ·

CARTO / Esri tiles

ARKESEL SMS

Ghanaian gateway, reached only from the Edge Function — the API key never ships in the app.

Cross-device flow: every write goes to Supabase; every client polls each ~20 s, diffs against what it has seen, and raises toasts, notifications, or the SOS alarm. The last snapshot is cached locally, so the feed renders offline and writes fall back to local-only.

Security model

- Passwords hashed server-side (Supabase Auth); sessions persist and refresh across devices.
- Row-level security: community posts — any signed-in user; admin bulletins, broadcasts, deletions — admins only (checked in the database); SOS events visible only to author, targets, and admins.
- The `send-sms` function authenticates every caller, caps recipients (5 for SOS, 50 for blasts), verifies admin role server-side for blasts, and normalizes Ghanaian numbers (`024...` → `23324...`).

STACK

Tech & data sources

LAYER	CHOICE
Framework	Expo SDK 54 · React Native 0.81 · TypeScript
Backend	Supabase — Auth, Postgres (RLS), Storage, Edge Functions
3D engine	MapLibre GL JS (WebGL)
Map tiles	CARTO Voyager & Dark Matter · Esri World Imagery — keyless
SMS	Arkesel (Ghana) via Edge Function; device composer fallback
Device APIs	expo-location · notifications · image-picker · sms · audio
Hosting	GitHub Pages (PWA) with self-update via <code>version.json</code>

DATA	SOURCE	COVERAGE
Disaster alerts	GDACS (UN)	Worldwide, incl. West Africa
Weather alerts	US National Weather Service	US only
Weather + forecast	Open-Meteo	Worldwide
Landmarks & search	OpenStreetMap	Ghana-first
Local Ghana alerts	SafeAlert admins & community	Ghana

DEVELOPMENT

Run & deploy

```
# local development
npm install
npm start           # Expo dev server → scan QR with Expo Go
npm run web         # browser preview

# checks
npx tsc --noEmit

# production
npm run deploy      # build + publish to GitHub Pages
```

Remote push notifications and app-store builds require an EAS development build; the poll + local-notification pattern delivers the same UX in the meantime.

Two-phone demo script

- 1 Phone A signs up with phone B's account email as an emergency contact.
- 2 Hold SOS on A → A shows "SMS delivered automatically (Arkesel)" and the call button.
- 3 Within ~20 s, phone B erupts in the full-screen siren alarm with B's one-tap map link.

HONEST NOTES

Known limitations — and the fix for each

LIMITATION	WHY	FIX WHEN NEEDED
~20 s delivery; no background push	Expo Go can't do FCM/APNs	EAS build + Expo Push
NWS alerts are US-only	No public Ghana Met API	GDACS + admin bulletins cover Ghana; add a GMet feed if one appears
SMS needs Arkesel credit	Trial balance exhausted by testing	Top up at sms.arkesel.com ; register the "SafeAlert" sender ID
Backend can pause	Supabase free tier sleeps after ~1 week idle	Restore in the Supabase dashboard (~2 min)
Web siren may wait for a tap	Browser autoplay policy	Native/Expo Go unaffected

Credentials: admin login details are provided separately and are not written in this document or the repository.

SafeAlert · built from the Group 11 project proposal · KNUST, Kumasi · full technical reference in [DOCUMENTATION.md](#) in the repository.